SafeSend

Safer crypto payments for people new to crypto, with family looking out.

Web3 for Social Good Hackathon 2026 · Tech Track · Live prototype on Monad testnet

The problem

S$410.6M	lost to scams in Singapore in the first half of 2026 (para 3).
S$65.5M	of that, about 16.0%, was cryptocurrency (para 11). The SPF notes scammers use crypto because "transactions are irreversible and difficult to trace".
S$42,347	average loss per elderly victim, the highest of any age group (para 17).

Source: Singapore Police Force, Mid-Year Scam and Cybercrime Brief 2026.

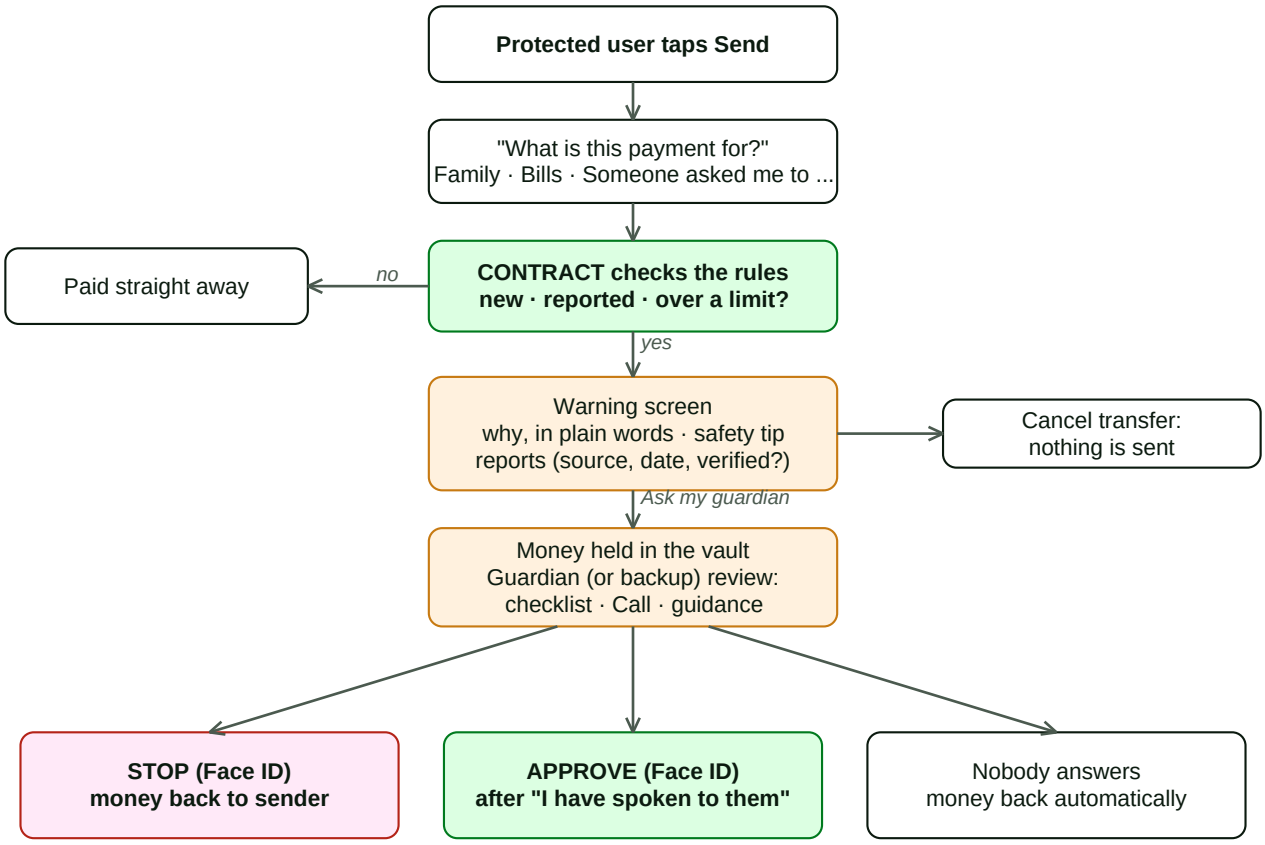
Safe and Coinbase already offer limits, delays, trusted contacts and recipient checks, but they are built for people who already understand crypto. The people who lose the most, older adults and teenagers with a first account, need the same protection in words they understand, with someone they trust beside them.

Who it is for

Protected user	Guardian
Mdm Tan, 68. A caller claiming to be a bank officer says her money must be moved to a "safe account".	**Wei Ming**, her son. Reviews any paused payment with a checklist; a backup guardian can step in.
A 16-year-old with a first crypto account, offered "guaranteed" returns in a group chat.	**Their parent.** Checks anything unusual before it leaves.

The solution

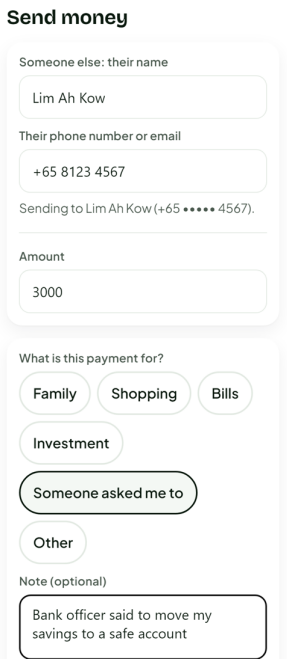
A smart contract pauses risky payments. The sender sees why in plain words. A trusted family member reviews it and decides with Face ID. **Golden rule:** contract rules and authorised people decide; AI only explains and can never move money.



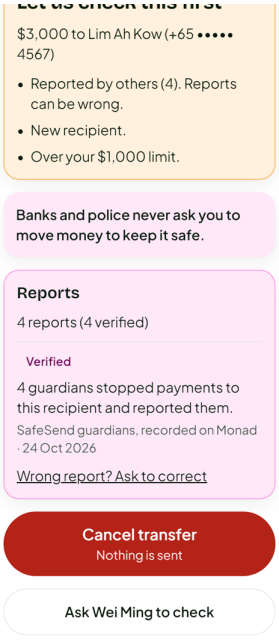
The contract pauses a payment when...	What the sender reads
the recipient has been reported	"Other people have reported this recipient (2 reports). Reports can be wrong, so your guardian checks."
the recipient is new	"You have not sent money to this person before."
one payment is over the limit	"This is more than the $1,000 limit your guardian set for one payment."
payments in a day or week add up past the total limit	"Together with your other payments recently, this goes over the total your guardian set."

The screens

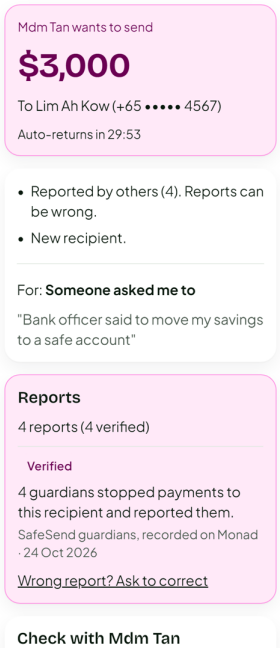
Real screenshots of the app, recorded on Monad testnet. Body text is 18px or larger, buttons are at least 56px tall, and there is a Large text switch.



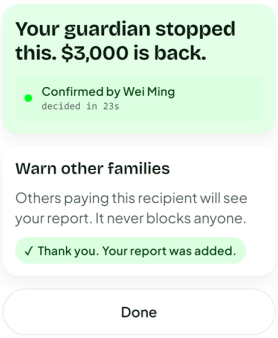
1 · What is it for?



2 · Paused, with reasons



3 · Guardian review



4 · Stopped, money back

Also built: the protected user can cancel a paused payment herself; an approved payment waits a 2-minute cooling-off during which she can still cancel; one-screen guardian setup with a daily total and a backup guardian (signed with Face ID); "No reports found" is never shown as "safe"; the guardian's Call button only uses the number they saved themselves; if nobody answers, the money returns automatically.

Why Web3

Property	Why it needs a blockchain
Nobody can override the rules	Pause rules and limits live in a smart contract. Our server only relays: it cannot release a payment, raise a limit or move money.
Only the guardian's Face ID releases money	The approval is a passkey signature verified inside the contract (P256). A server or stolen password cannot fake it.
The fail-safe does not depend on us	Anyone can trigger an expiry, which returns money to the sender. Funds are never stuck, even if SafeSend shut down.
Reports are a public good	Scam reports live in a public contract that any wallet or exchange can read. Each one is tied to a real stopped payment.
Auditable	Every pause, stop, approval and expiry is a public event.

Social impact

SafeSend interrupts the moment scams rely on: a person alone, under pressure, making an irreversible transfer. It asks every sender of an unusual payment "What is this payment for?". "Someone asked me to" is the clearest single scam signal, and it reaches the guardian even when no reports exist. Honest reporting, where no reports never means safe, avoids a false all-clear on a brand-new scam address.

Feasibility and sustainability

Stage	Plan
1. Prototype	Done: full journey on Monad testnet.
2. Pilot	10–20 families via SMU and community groups; measure stop rates and review times.
3. Mainnet	Real stablecoins after a security audit of the contract.
4. Embed	Wallets and exchanges add the pause-and-review flow to their apps and pay for it; families use it free; grants fund the pilot and audit.

How it was built and tested

Part	Details
Smart contracts	Solidity (Foundry). Pause rules, rolling total limit, backup guardian, sender cancel, cooling-off after approval, onchain passkey checks. 126 tests passing.
Network	Monad testnet (EVM). Vault 0x918022A5aDd2d250617ec64F7b38C5EEcf28E48f · Reports 0x3b7e4d3B44E91d2C0396d781F2BcdfE5fa87Cd48
App	Expo / React Native web app with Face ID passkeys.
Relayer and agent	The relayer pays network fees and holds no spending power. The agent writes guidance with an AI model, or with fixed rules when no model is available.
End-to-end test	The full journey (setup, payment, scam paused and stopped, unanswered pause refunded) passes on Monad testnet.

AI guidance: evaluation

Twenty scenarios: ten used once to tune the AI prompt, ten fresh ones written afterwards and never used for tuning. Each payment was already paused by the contract; the question is whether the guidance warns on scams without alarming on genuine payments.

Guidance	Scams warned (all 20)	False alarms (all 20)	Scams warned (fresh 10)
Fixed rules (fallback)	7 / 10	3 / 10	2 / 5
AI (gpt-oss-120b, free on Groq)	10 / 10	3 / 10	5 / 5

The AI catches scams with no obvious keywords that the rules miss. Both still raise false alarms on genuine payments that mention police or banks; the guardian decides either way.

Known limits

Testnet only, with stand-in dollars. Reports come only from SafeSend guardians who stopped a payment; an external source such as Chainabuse is not yet connected. Ten scenarios is a sanity check, not a study.

Built on open source

SafeSend extends Companion (github.com/SamuelDharshi/Companion, MIT licence), reusing its vault, payment-hold and guardian-passkey design. New in SafeSend: the Singapore scam-prevention focus, rolling total limits, backup guardian, plain-language warnings, the payment-purpose question, onchain reports shown honestly, the guided guardian review, AI guidance with evaluation, the elder-friendly UI and serverless hosting.